

PHYS40181

Hidden Forces, Learned Solutions

Theo Broxton

This project was carried out in collaboration with Erin Wright under the supervision of Christoph Englert and Pete Millington.

theo.broxton@student.manchester.ac.uk

ABSTRACT: We investigate higher-dimension operators in the Vainshtein regime of an infra-red massive Galileon effective field theory. Using homotopy continuation within the φ enics finite-element solver, converged solutions containing both $n = 3$ and $n = 4$ operators are obtained in regimes where direct solution methods previously failed. The resulting field profiles exhibit large shifts in the Vainshtein radius, suggesting that higher-order operators may not remain perturbatively suppressed in the screened regime. Neural-network differential equation solvers are also explored using **Elvet**. Whilst the framework successfully reproduces linear benchmark equations, the full nonlinear Galileon system proves substantially more challenging, highlighting current limitations of neural-network solvers for strongly nonlinear systems with widely separated physical scales.

Contents

1	Introduction	1
2	Background & Theory	3
2.1	The IR Galileon and Vainshtein screening	3
2.2	Continuation methods	4
2.3	Neural networks	6
2.3.1	The Elvet solver	8
3	φenics continued	9
4	Elvet	10
4.1	The Klein-Gordon equation	10
4.2	The IR equation of motion	12
5	Discussion & Outlook	15
	References	17

1 Introduction

Gravity is the dominant force shaping the large-scale structure of the Universe, yet it remains in many respects the least well understood of the fundamental interactions. Our best description is Einstein’s theory of General Relativity (GR), characterised by the great physicist Dirac as ‘probably the greatest scientific discovery ever made’ [1]. The theoretical leap to supersede Newtonian action at a distance with spacetime curvature was undoubtedly brilliant, and is expressed succinctly in the Einstein field equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (1.1)$$

where $G_{\mu\nu}$ is the Einstein tensor encoding curvature, G is Newton’s gravitational constant, c is the speed of light, and $T_{\mu\nu}$ is the energy-momentum tensor. The phenomenology of the theory has developed to span black hole thermodynamics [2] to modern cosmology [3].

In the century since its origination, the theory has been extensively tested. This ranges from higher precision measurements [4, 5] of Einstein’s original tests [6] (anomalous perihelion precession of Mercury, gravitational redshift and deflection of light¹) to measurement of the Shapiro time delay [9, 10] and detection of gravitational waves with LIGO [11].

It is perhaps surprising then that, despite its many triumphs, there remain compelling reasons to suspect that GR is not the final word on gravity. GR is widely understood to be an effective field theory (EFT) valid only below the Planck scale M_{Pl} [12], expected to receive corrections from a more complete description at higher energies – much as Fermi’s theory of weak interactions gives way to the full electroweak theory at the scale of the W and Z bosons [13]. *Modified gravity* theories, often expressed as alterations of the Einstein-Hilbert action, are required to reduce to GR in an appropriate limit, in much the same way that Einstein’s theory reduces to Newtonian gravity at low energies. A systematic framework for comparing such theories with experiment is provided by the Parameterised Post-Newtonian (PPN) formalism [14]. The landscape of modified gravity theories is broad, encompassing higher-dimensional scenarios such as Kaluza-Klein models [15, 16], braneworld constructions [17, 18], as well as scalar-tensor theories including Brans-Dicke gravity [19] and $f(R)$ theories.

One observational motivation for departing from Einstein’s theory is the accelerated expansion of the Universe. The Λ -cold dark matter (Λ CDM) model, supported by measurements of type Ia supernovae and the Cosmic Microwave Background (CMB) [20], has a problematic cosmological constant (CC) Λ . There is a discrepancy of around 60 orders between what we measure ($\Lambda_{\text{obs}} \sim 10^{-120} M_{\text{Pl}}^4$) and

¹This was famously originally measured by Eddington [7, 8].

predict² ($\Lambda_{\text{theory}} \sim 10^{-60} M_{\text{Pl}}^4$), which poses a severe naturalness problem. While an anthropic argument could be made, alternative dynamical solutions are more appealing.

A natural class of candidates introduces an additional scalar degree of freedom coupled to matter. However, any such modification must reproduce the well-tested predictions of GR locally, for instance within the solar system. Rather than achieving this through unnaturally small couplings, a more theoretically appealing route is provided by screening mechanisms, which can suppress the effect of the scalar field in regions where GR is precisely tested [22]. Here, nonlinear derivative self-interactions of the scalar field, termed the *Vainshtein mechanism*, are considered³. The realisation of this mechanism that we study is an infra-red (IR) Galileon [24] EFT, whose equation of motion (eom) is

$$\square\pi - m_\pi^2\pi - \epsilon \frac{\square((\square\pi)^n)}{\Lambda^{3n-1}} = \frac{\rho}{M_{\text{Pl}}}, \quad (1.3)$$

where m_π is the scalar field mass, Λ is the EFT cutoff scale and ϵ controls the strength of the nonlinear interaction. This was studied numerically using the *varnics* finite element solver [25] by Braden et al. [26]. For a valid EFT description, higher-dimension operators – corresponding to larger values of n – should be progressively suppressed by powers of the cutoff Λ . Whether this hierarchy is actually respected in the Vainshtein regime, where the nonlinear interactions are by definition large, is a non-trivial question.

In semester 7 of the project, this question was investigated by attempting to solve the eom with both $n = 3$ and $n = 4$ operators included simultaneously [27]. The *varnics* solver was found to fail to converge when both operators were present, preventing a physical conclusion from being drawn. Independent solutions for each operator suggested that higher-dimension operators may compete with or dominate lower-dimension ones in the Vainshtein regime (contrary to EFT expectations), but a definitive statement could not be made on the health of the EFT.

This report shifts the focus to numerical analysis of the eom, and two complementary approaches are investigated. The first employs continuation methods, in which the higher-order operator is introduced gradually using a homotopy parameter, allowing converged solutions from one parameter value to seed subsequent

²Interpreted as the energy density of the vacuum, we can estimate its size as [21],

$$\langle\rho\rangle\sim\int_0^{\Lambda_{\text{UV}}}dk\,k^2\sqrt{k^2+m^2}\sim\Lambda_{\text{UV}}^4, \quad (1.2)$$

where we have simply summed zero-point energies of Standard Model (SM) fields up to an ultraviolet cutoff Λ_{UV} . Following the rationale of [22] that the theory is tested up to the weak scale, we take $\Lambda_{\text{UV}} \sim 1$ TeV.

³Other screening mechanisms include symmetron and chameleon models [23].

ones. Continuation techniques are widely used in nonlinear differential equations and bifurcation theory near difficult regions of parameter space [28].

The second approach explores an alternative class of differential equation solvers based on neural networks. Specifically, we employ the **Elvet** framework [29], which represents the field solution as a trainable neural network. This is determined by minimising a loss functional constructed from the differential equation residual and boundary conditions. Unlike finite-element discretisations, this approach operates directly on a continuous representation of the solution and provides an independent numerical framework against which the φ enics results may be compared.

The report is structured as follows. Section 2 briefly reviews the theoretical background of the IR Galileon and Vainshtein screening, and introduces the frameworks underpinning continuation methods and neural-network differential equation solvers. Section 3 describes the implementation and results of the continuation approach within φ enics, including the combined $n = 3$, $n = 4$ solutions. Section 4 presents the **Elvet** implementation, beginning with validation on the Klein-Gordon (KG) equation before proceeding to the full IR eom. Section 5 draws conclusions from both approaches and discusses directions for future work.

2 Background & Theory

2.1 The IR Galileon and Vainshtein screening

Since the theoretical motivation and formalism underpinning this project were developed in detail in the semester 7 report [27], we restrict ourselves here to a concise summary of the key ideas.

The modification of gravity considered in this project is the introduction of an additional scalar degree of freedom π , coupled to matter. Such a scalar field generically mediates a long-range fifth force, which would produce observable deviations from GR on solar-system scales where the theory has stringent constraints. The Vainshtein mechanism suppresses the scalar-mediated force within a characteristic *Vainshtein radius* r_V . Outside this radius, the scalar force is unsuppressed, potentially allowing for the significant modifications to gravity on large scales, whilst inside it GR is recovered. Consult our previous work [27] for an outline of how this arises with an effective kinetic coefficient $Z_{\mu\nu}$.

The scalar theories we consider are Galileons, defined by the requirement that the Lagrangian be invariant under the internal symmetry [24]

$$\pi(x) \rightarrow \pi(x) + b_\mu x^\mu + c, \quad (2.1)$$

where c is a constant shift and b_μ characterises a constant gradient shift. This symmetry strongly constrains the allowed interaction terms. When one also requires that the eom be second-order to avoid the Ostrogradsky instability [30], there are

only $(d + 1)$ independent Galileon Lagrangians in d dimensions of the form $\mathcal{L}_n \sim (\partial^2 \pi)^{n-2} \partial \pi \partial \pi$. Galileon theories are known to exhibit Vainshtein screening through their nonlinear derivative structure, making them a natural arena in which to study this mechanism. Due to the inherent nonlinearity, numerical methods are typically necessary however a sketch is given in our previous work [27] (based upon [22, 31]) to analytically show Vainshtein screening in the cubic Galileon theory.

In this project we work with an IR massive Galileon theory, which can be understood as arising from integrating out a heavy scalar field H from a UV-complete massive Galileon action [32]. Following the treatment in Burrage et al. [33] (and described in [27]), this procedure yields a low-energy effective action for the light field π alone, with eom

$$\nabla_r^2 \pi - m_\pi^2 \pi - \epsilon \frac{\nabla_r^2 ((\nabla_r^2 \pi)^n)}{\Lambda^{3n-1}} = \frac{\rho}{M_{\text{Pl}}}, \quad (2.2)$$

where spherical symmetry has been assumed and parameters are as in Equation 1.3. The nonlinear term is responsible for Vainshtein screening: in the vicinity of a massive source, derivatives of the field become large and this term dominates, suppressing the scalar force. Far from the source, the field is well described by the linear massive KG equation and decays following a Yukawa profile $\sim e^{-m_\pi r}/r$.

As alluded to in section 1, operators of progressively higher dimension should remain suppressed by powers of the cutoff scale Λ in a consistent EFT. Consequently, lower-order operators are expected to dominate the dynamics below the cutoff, whilst higher-order operators only provide perturbative modifications.

2.2 Continuation methods

Continuation methods are numerical techniques designed to improve the robustness of solving nonlinear equations by introducing difficult terms gradually. Rather than attempting to solve a highly nonlinear system directly, one instead constructs a continuous family of related problems parameterised by some quantity λ , chosen such that the system is easily solvable for an initial value λ_0 . The solution is then tracked as the parameter is varied toward the desired target value. This is a well-developed branch of numerical mathematics; a thorough treatment can be found in [28, 34].

Schematically, one considers a nonlinear problem of the form

$$F(u, \lambda) = 0, \quad (2.3)$$

where u denotes the unknown solution and λ is the continuation parameter. The set of solutions may then be viewed geometrically as a curve in the (u, λ) parameter space. This may in general contain turning points – values of λ at which the solution branch folds back – and bifurcations, where multiple solution branches intersect.

If a solution $u(\lambda_0)$ is known for some initial parameter value λ_0 , it can be used as the initial guess for a neighbouring parameter value $\lambda_0 + \Delta\lambda$. Repeating this process iteratively allows one to follow a branch of solutions through parameter space, provided the step size $\Delta\lambda$ is chosen small enough that successive iterations remain close.

The motivation for this approach can be understood from the behaviour of non-linear iterative solvers such as Newton-Raphson methods. These methods typically converge only when the initial guess lies sufficiently close to the true solution. For strongly nonlinear systems, directly solving the full equations may fail because a suitable starting point is not known a priori, causing the iteration to diverge or converge toward an unphysical branch of solutions. Continuation methods address this by ensuring that each initial guess is the converged solution of a nearby problem, so that successive solutions differ only slightly.

A common implementation is *homotopy continuation* [28], in which one deforms a simple problem continuously into the full nonlinear problem of interest. If

$$F(u) = L(u) + N(u), \quad (2.4)$$

where $L(u)$ is a linear contribution and $N(u)$ contains nonlinear terms, one may instead solve the homotopy

$$H(u, \lambda) = L(u) + \lambda N(u) = 0 \quad (2.5)$$

for increasing values of $\lambda \in [0, 1]$. The problem at $\lambda = 0$ is typically much easier to solve, whilst $\lambda = 1$ recovers the full equations. If the step size $\Delta\lambda$ is sufficiently small, the converged solution from one step generally lies within the basin of convergence for the next⁴.

In the context of the IR Galileon eom, the continuation parameter controls the relative strength of the $n = 4$ nonlinear operator. Writing the equation schematically as

$$H(u, \alpha, \beta) = L(u) + \alpha D_3(u) + \beta D_4(u) = 0, \quad (2.6)$$

where D_n denotes the nonlinear operator of order n , one begins at $(\alpha, \beta) = (1, 0)$ for the converged $n = 3$ solution, and then gradually increases β towards its physical value. The aim is to determine whether the convergence failures encountered in the direct solve reflect a genuine absence of solutions in the combined system, or merely the sensitivity of the solver to its starting point.

⁴More sophisticated continuation strategies exist for cases where simple parameter continuation fails. Pseudo-arclength continuation, for instance, parameterises the solution curve by arc length rather than by λ directly, allowing one to continue through turning points where $du/d\lambda$ diverges [28]. Such methods are widely used in bifurcation theory and the study of dynamical systems.

2.3 Neural networks

An alternative approach to solving differential equations, which has gained significant traction in recent years [35], is to represent the solution as the output of a trained neural network (NN). The significance of NNs to modern science was underscored in 2024, when Geoffrey Hinton and John Hopfield were awarded the Nobel Prize in Physics [36] for their foundational contributions to artificial neural networks⁵. Rather than discretising the domain into a finite set of points and solving for the values of the solution at each, this approach seeks a continuous function defined over the entire domain, parameterised by the weights and biases of the network.

A NN is a parametric model constructed by composing layers of simple nonlinear functions. In the simplest case, a single layer maps an input vector $x \in \mathbb{R}^M$ to an output vector $a \in \mathbb{R}^J$ via an affine transformation followed by a componentwise nonlinearity,

$$a = \varphi(Wx + b), \quad (2.7)$$

where $W \in \mathbb{R}^{J \times M}$ is a weight matrix, $b \in \mathbb{R}^J$ is a bias vector, and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is an activation function applied componentwise [37]. Such a mapping is called a single layer perceptron. Common choices of activation function include the logistic (sigmoid) function $\varphi(x) = (1 + e^{-x})^{-1}$, which provides a smooth interpolation between 0 and 1, and the rectified linear unit (ReLU) $\varphi(x) = \max\{x, 0\}$. A multilayer perceptron, or *deep neural network*, is formed by composing L such layers sequentially,

$$H(x; (W^{(\ell)}, b^{(\ell)})_{\ell=1}^L) = a^{(L)}, \quad a^{(\ell)} = \varphi(W^{(\ell)}a^{(\ell-1)} + b^{(\ell)}), \quad a^{(0)} = x, \quad (2.8)$$

where the intermediate quantities $a^{(\ell)}$ are referred to as hidden layers and $a^{(L)}$ as the output layer. See Figure 1 for a schematic illustration of this iterative process. The total number of layers is called the depth of a network and the dimension of each layer's output is its width (e.g. $a^{(2)} \in \mathbb{R}^m$ in Figure 1 has width m). The resulting network defines a highly flexible nonlinear function whose expressive power generally increases with both network depth and width.

A theoretical justification for using neural networks as function approximators is provided by the Universal Approximation Theorem [39]. In one common form, the theorem states that two-layer networks with non-polynomial activation functions are dense in the space of continuous functions on compact domains⁶ [40]. Consequently, sufficiently wide neural networks can approximate arbitrary continuous functions to arbitrary accuracy. In practice, greater depth is still important for a fixed parameter budget; Telgarsky [41] showed this when considering the asymptotic number of neurons required for networks of different depths to represent the same function.

⁵The award of the physics prize to Hinton and Hopfield for their contributions to neural networks was somewhat controversial, although certainly inspired by the physics of statistical mechanics.

⁶Generalisations exist for discontinuous functions and non-compact domains such as $\mathcal{X} = \mathbb{R}^M$, though these require a weaker notion of density [37].

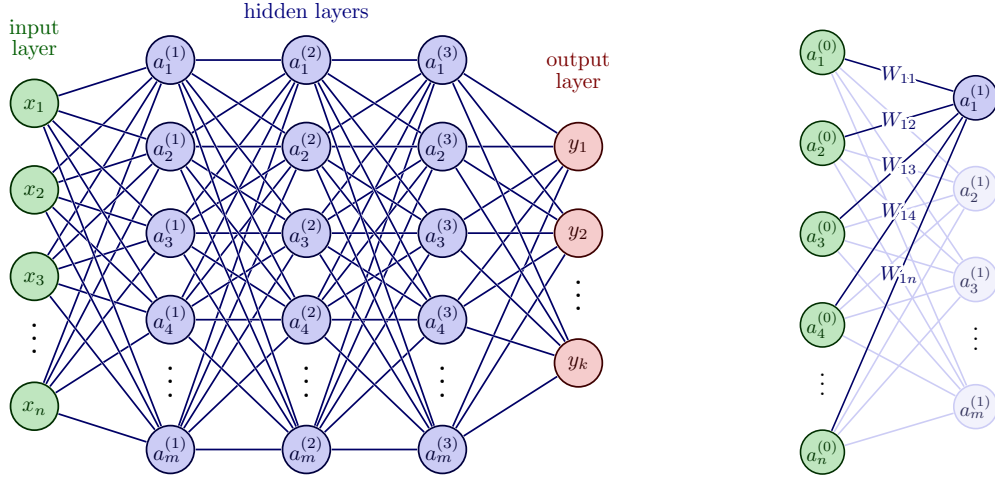


Figure 1: NN diagrams adapted from [38]. *Left*: diagram showing layers and connections of a neural network with architecture (n, m, m, m, k) , where each entry denotes the width of that layer. The network can thus be viewed as a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^k$. *Right*: schematic illustrating how elements of the weight matrix W are combined to form the first component of the $a^{(1)}$ vector, $a_1^{(1)} = \varphi \left(\sum_i W_{1i} a_i^{(0)} + b_1^{(1)} \right)$.

The parameters of the network – the weights and biases across all layers – are determined by minimising a *loss function* $\mathcal{L}$, which is ubiquitous in machine learning⁷. A loss function is constructed such that a network achieving a small loss value approximately solves the problem of interest.

A common loss function is the quadratic or squared loss

$$\mathcal{L}_t(y) = \|y - y^{(t)}\|^2, \quad (2.9)$$

where the values $y^{(t)}$ at $x^{(t)}$ are known⁸. Clearly, the larger the deviation of y either above or below the known value, the greater the loss. Losses can also be modified by the addition of penalty terms known as regularisation.

The process of minimisation is known as *training* a NN, often achieved iteratively through gradient descent. Starting from an initial parameter vector θ_0 , successive updates take the form

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t), \quad (2.10)$$

where $\eta > 0$ is the learning rate [42]. Subtracting a step in the direction of the gradient moves the parameters towards a local minimum of the loss. The required gradients are computed by *back propagation*, which applies the chain rule in reverse order through the network, reusing intermediate quantities already evaluated in the forward pass [42]. In practice, stochastic variants are preferred - these approximate

⁷Loss functions have actually been around for centuries, used in linear regression for example.

⁸Examples of other loss functions are the cross-entropy loss, 0-1 loss and L^1 loss.

the full gradient using only a single randomly chosen point per step, which is far cheaper when the training set is large. **Elvet** uses the **Adam** optimiser [43], which extends stochastic gradient descent using adaptive per-parameter learning rates. See [44] for a review of optimisation methods.

2.3.1 The Elvet solver

To solve differential equations, **Elvet** [29] represents the solution as a fully connected neural network. This is trained with a loss function constructed from the residuals of the differential equation and boundary conditions evaluated at sampled training points⁹. We format our equations and boundary conditions as

$$\mathcal{E}_k(x)[f] = 0, \quad \mathcal{B}_l(x)[f] = 0, \quad (2.12)$$

where these are local functionals of the form

$$\mathcal{E}(x)[f] = E(x_i, y_i, \partial_i y_j, \partial_i \partial_j y_k, \dots), \quad (2.13)$$

$$\mathcal{B}(x)[f] = B(x_i, y_i, \partial_i y_j, \partial_i \partial_j y_k, \dots), \quad (2.14)$$

and $y = f(x)$. The loss density

$$\hat{\mathcal{L}}(x) = \sum_k \mathcal{E}_k(x)^2 + \sum_l \mathcal{B}_l(x)^2, \quad (2.15)$$

is a local measure of the loss which is then summed over training points $x^{(t)}$ to yield the loss,

$$\mathcal{L}[f] = \sum_t \hat{\mathcal{L}}(x^{(t)})[f]. \quad (2.16)$$

The loss and its gradients are calculated using the Python library **TensorFlow** [45] to train the model, in a process that occurs in repeated steps called epochs. For further details on the implementation, consult the **Elvet** paper [29] and documentation [46].

Unlike mesh-based methods, the resulting solution is represented by a continuous differentiable function defined throughout the domain. Refinement may therefore be achieved either by increasing the network capacity or by sampling additional collocation points during training.

⁹Differential equations are treated as the degenerate case of functional minimisation of a functional $\mathcal{F}$ subject to constraints $\mathcal{C}[f] = 0$, with loss

$$\mathcal{L}[f] = \mathcal{F}[f] + W\mathcal{C}[f]^2, \quad (2.11)$$

where W is some hyperweight [29].

3 φ enics continued

As described in Equation 2.6, continuation is initiated from the converged $n = 3$ solution at $\beta = 0$, with β subsequently incremented until $\beta = 1$, where both nonlinear operators are entirely present simultaneously.

The continuation procedure was found to be sensitive to several numerical parameters. Solution convergence depended on the internal scaling of the φ enics solver [25], and care was required in choosing the step size $\Delta\beta$; steps too large caused the Newton-Raphson iteration to fail, whilst smaller steps dramatically increased computational cost. Residual numerical noise was observed in certain iterations, though this was substantially reduced compared with the direct solve without continuation.

In Figure 2, we present field solutions near the source for increasing values of β , alongside a comparison of the $\beta = 0$ and $\beta = 1$ solutions across the full domain. As β increases, the field magnitude $|\pi|$ is progressively reduced near the source. More significantly, the introduction of the higher-order operator D_4 shifts the Vainshtein radius r_V outward by more than an order of magnitude relative to the $n = 3$ solution alone. This represents a qualitative modification of the field profile rather than a small perturbative correction, which is in tension with naive EFT expectations [13].

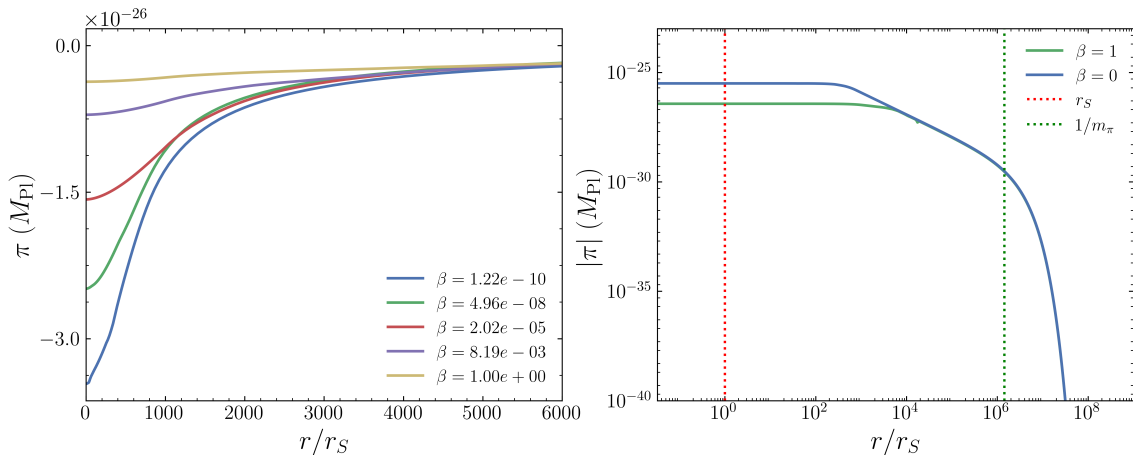


Figure 2: The left plot show the near-source field behaviour for increasing values of β . The field magnitude clearly becomes progressively smaller. The right plot shows the $\beta = 0$ and $\beta = 1$ solutions across the domain, with separate Vainshtein radii and a two stage gradient change for $\beta = 1$. Here, the continuation parameter β was iterated 24 times. r_S is the source (smoothed top hat) radius and $1/m_\pi$ is the Compton wavelength.

An additional feature is the emergence of a two-stage power-law behaviour in the $\beta = 1$ solution: a region of shallower gradient at $r \sim 10^3$ transitions into a steeper descent at $r \sim 10^4$. We attribute this to the competing dynamics of the D_3 and D_4 operators, which dominate at different scales. A force ratio plot analogous to Figure 2 of [27] confirms that a screening regime remains present, though it is accompanied

by numerical noise towards the end of the screening. This may reflect amplification of solution-level noise through field differentiation, or may signal genuine numerical instability associated with the broadened Vainshtein radius. Further development of the continuation results, including a more systematic study of step-size dependence and scaling, is presented in [47].

4 Elvet

Having outlined the theoretical framework underlying neural-network differential equation solvers in subsection 2.3, we now consider the practical implementation of **Elvet**. The framework has previously been demonstrated on a range of problems including ordinary differential equations, the quantum harmonic oscillator, Lotka–Volterra systems and simple partial differential equations [29]. We begin by producing some simpler results in order to benchmark the solver and gain familiarity with its behaviour before considering the full IR Galileon equations of motion.

4.1 The Klein-Gordon equation

The eom, Equation 2.2, is composed of a nonlinear part and a linear part – the Klein-Gordon (KG) equation. We first consider the homogeneous KG equation,

$$(\square + m^2) \phi(\mathbf{x}, t) = 0. \quad (4.1)$$

In the static, spherically symmetric case, the d’Alembertian reduces to the radial Laplacian,

$$\nabla_r^2 \phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \frac{d^2 \phi}{dr^2} + \frac{2}{r} \frac{d\phi}{dr}, \quad (4.2)$$

so the KG equation becomes

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - m^2 \right) \phi(r) = 0, \quad (4.3)$$

where we use ϕ here to disambiguate from the Galileon field.

To implement the problem in **Elvet**, we first construct an **Elvet box** — a `tf.Tensor` defining the collocation points over the radial domain — which is supplied as the `domain` argument to `elvet.solver`. We use 500 linearly spaced points in the interval $(r_{\min}, r_{\max}) = (10^{-1}, 10)$ and fix the mass parameter to $m = 1$. The boundary conditions imposed are¹⁰

$$\phi(r_{\min}) = 1, \quad \phi(r_{\max}) = 0, \quad (4.4)$$

which select the decaying Yukawa-like solution and remove the exponentially growing mode $\sim e^{mr}$.

¹⁰In **Elvet**, boundary conditions are specified as pairs of the form (point, equation = 0) [46].

The differential equation itself is implemented in **Elvet**’s generic form using the arguments (`domain`, `*derivatives`). For the Klein–Gordon equation, this corresponds to constructing a function with inputs (`r`, `phi`, `dphi`, `d2phi`) representing the field and its derivatives. We then define a fully connected neural network using `elvet.nn` with architecture (1, 10, 1), corresponding to a single hidden layer of width 10 between one input neuron r and one output neuron ϕ (see Figure 1). The network is trained for 50 000 epochs, with the full optimisation taking less than one minute on a standard hardware.

In the original formulation, training rapidly plateaued at an approximately constant loss value whilst producing a solution inconsistent with the expected Yukawa decay. This behaviour is likely associated with the singular coefficient $2/r$, which becomes large near the origin and may destabilise optimisation despite the finite cutoff $r_{\min} > 0$.

An alternative approach is via the substitution

$$u(r) = r\phi(r) \longrightarrow u'' - m^2u = 0, \quad (4.5)$$

where $'$ denotes differentiation with respect to r . This transformation removes the first-derivative term arising from spherical geometry, reducing the equation to a one-dimensional Helmholtz-type equation. The original ϕ can then be retrieved using the substitution relation.

Here we will present the solution found after multiplying the equation through by r^2 , yielding

$$r^2\phi'' + 2r\phi' - m^2r^2\phi = 0. \quad (4.6)$$

This removes the explicit singular coefficient and substantially improves optimisation stability. The resulting **Elvet** solution is compared in Figure 3 with an independent numerical solution obtained using Mathematica’s **NDSolve**. The two are in excellent agreement across the domain, with the residual difference shown in the middle panel at the level of $\mathcal{O}(10^{-3})$.

In addition, the loss density $\hat{\mathcal{L}}$ is shown in the lower panel of Figure 3. This is largest near $r_{\min}$, where the solution varies most rapidly and the optimisation must simultaneously satisfy both the differential equation and imposed boundary conditions. A smaller increase is also visible near $r_{\max}$, associated with enforcement of the asymptotic boundary condition. Overall, the low loss density and strong agreement with **NDSolve** provide a useful validation benchmark for subsequent applications of **Elvet** to more complicated nonlinear equations.

The natural next step was a simplified version of the full IR Galileon equation without nested Laplacians. While an **Elvet** solution was obtained, independent verification proved more difficult in this case as Mathematica’s **NDSolve** failed to converge reliably for the parameter choices investigated.

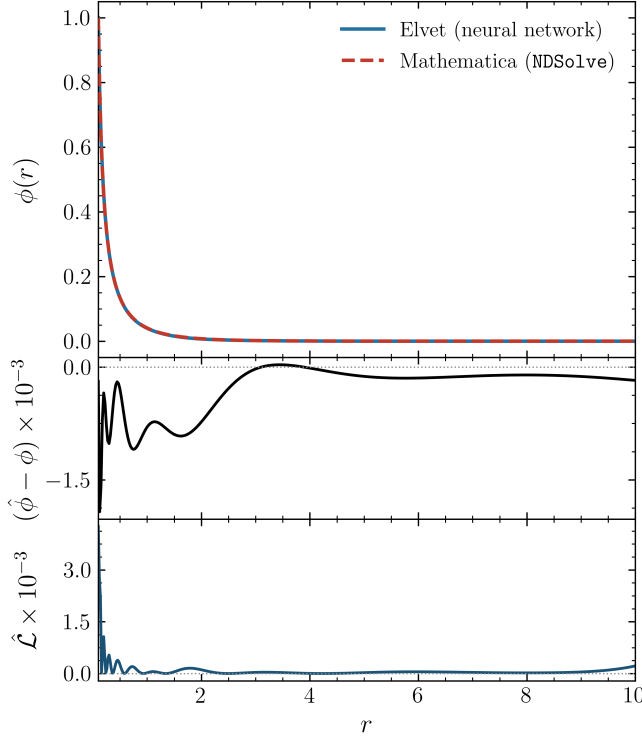


Figure 3: The upper panel shows the **Elvet** (solid blue) and Mathematica (dashed red) solutions to the KG equation with boundary conditions and parameters as given in the text. The middle panel shows the difference between the two predicted solutions and the lower panel shows the loss density $\hat{\mathcal{L}}$. We observe that the loss density and relative difference between solutions is greatest near $r_{\min}$.

4.2 The IR equation of motion

Having validated **Elvet** on the linear KG equation and solved a simplified nonlinear equation, we then attempted to solve the full nonlinear IR Galileon equation. Specifically, the $n = 3$ case was considered, which for concreteness is

$$\nabla_r^2 \pi - m_\pi^2 \pi - \epsilon \frac{\nabla_r^2 ((\nabla_r^2 \pi)^3)}{\Lambda^8} = \frac{\rho}{M_{\text{Pl}}} . \quad (4.7)$$

Direct implementation of the nested Laplacian operator proved numerically unstable, so the equation was reformulated as a coupled system of lower-order equations following the same auxiliary variable construction used in φ enics [26]. These are

$$Y - \nabla_r^2 \pi = 0 , \quad (4.8)$$

$$W - Y^n = 0 , \quad (4.9)$$

$$Y - m_\pi^2 \pi - \epsilon \frac{\nabla_r^2 W}{\Lambda^{3n-1}} - \frac{\rho}{M_{\text{Pl}}} = 0 , \quad (4.10)$$

where the first two equations define the auxiliary variables Y and W . The neural network therefore outputs the three functions (π, Y, W) simultaneously. Boundary conditions for the problem were taken as [26]

$$\pi(\infty) = 0, \quad \frac{\partial}{\partial r} \pi(0) = 0, \quad [(\nabla_r^2 \pi)^n](\infty) = 0, \quad \frac{\partial}{\partial r} [(\nabla_r^2 \pi)^n](0) < \infty . \quad (4.11)$$

The implementation presented several significant challenges. The equations were rescaled to an $\mathcal{O}(1)$ form, following the same procedure as in [27], since the physical parameter set spans many orders of magnitude¹¹.

Experimentation was carried out across a range of network architectures, with $(1, 20, 20, 20, 3)$ serving as a representative choice balancing expressive capacity against training time. Whilst **Elvet** consistently reduced the total loss over training, the resulting field profiles were found to be highly sensitive to both the network architecture and the random initialisation of weights. Increasing either the depth or width of the network altered the qualitative behaviour of the solutions substantially, and repeated runs with identical hyperparameters frequently converged toward visibly different field profiles despite achieving comparable loss values. This behaviour is indicative of a highly non-convex optimisation landscape containing multiple local minima, possibly corresponding to distinct physical or unphysical solution branches.

A further difficulty arose from the enormous hierarchy of scales intrinsic to the problem. Physically relevant solutions require simultaneously resolving the strongly nonlinear Vainshtein region near the source together with the exponentially suppressed Yukawa decay at large radii, where $\pi(r) \sim e^{-m_\pi r}/r$ reaches values of order 10^{-40} or smaller. Consequently, the contribution of this region to the total loss is negligible compared with the nonlinear inner region, meaning that optimisation has little incentive to reproduce the correct asymptotic decay.

This issue was reflected consistently in the obtained solutions. Although the neural network frequently captured nontrivial behaviour near the source, the field generally failed to exhibit the expected Yukawa fall-off at large r . Instead, the solutions tended to asymptote toward approximately constant values on log-log plots.

Various strategies were explored to address this. The collocation point distribution was changed from linear to logarithmic spacing, substantially improving resolution in the small- r region. Boundary conditions derived from corresponding φ enics solutions were incorporated, including conditions on π , Y , and derivatives of W at $r_{\min}$. A Robin boundary condition motivated by the asymptotic Yukawa solution was also imposed, replacing the simpler Dirichlet condition $\pi(r_{\max}) = 0$. Neglecting nonlinear terms at large r and differentiating the decaying solution yields

$$\frac{d\pi}{dr} + \left(m_\pi + \frac{1}{r}\right)\pi = 0, \quad (4.12)$$

which encodes the exponential decay. Despite this physically motivated modification, the network did not reliably recover stable Yukawa-like behaviour.

Motivated by the exponential suppression, a reweighting of the loss density was also explored, multiplying residual terms by factors $\propto e^{m_\pi r}$ to amplify the contribution of large- r collocation points. Whilst this altered the course of optimisation,

¹¹Take for example $\Lambda = 10^{-46}$ raised to the power of 8 in the denominator of a term ...

it did not produce consistently physical solutions. Efforts to separately learn the small- and large- r regimes through domain decomposition similarly did not yield the desired results.

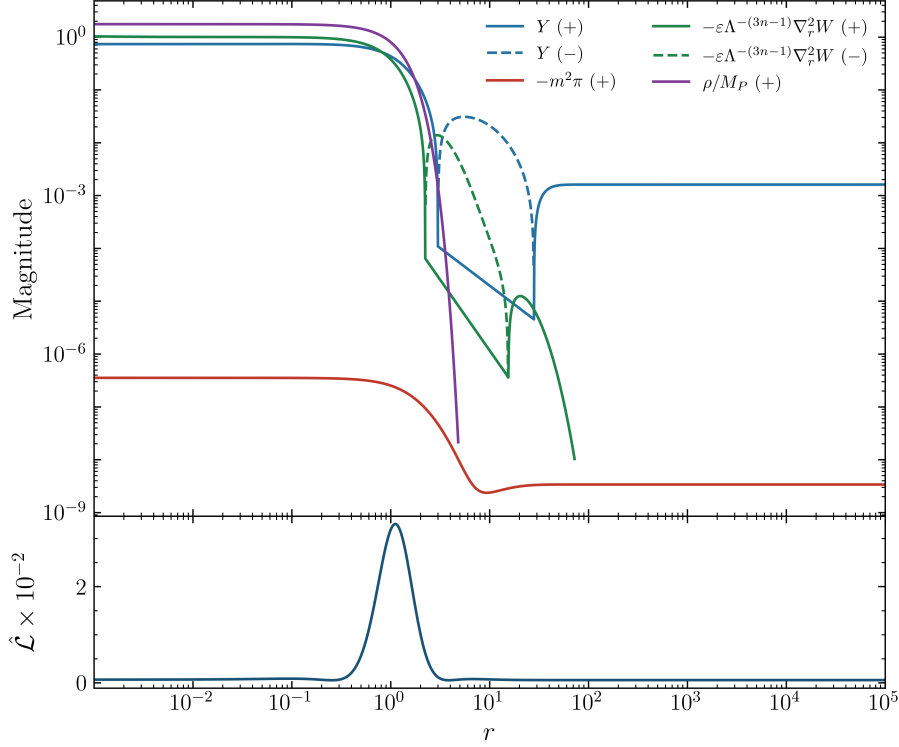


Figure 4: The upper panel shows the individual contributions to the rescaled eom obtained using **Elvet**. The nonlinear hierarchy is qualitatively reproduced near the source, although the field fails to transition into the expected Yukawa regime at large radius. The lower panel shows the loss density, which peaks near $r \sim 1$, suggesting that optimisation difficulty is concentrated near the transition between screened and asymptotic behaviour.

The term-by-term contributions to the rescaled eom are shown in Figure 4. Whilst the hierarchy of terms near the source is correctly reproduced, the field fails to transition into the expected Yukawa decay beyond the source, instead plateauing at large r . The loss density peaks near $r \sim 1$, indicating the optimisation encounters the greatest difficulty in the transitional regime. Figure 5 shows the near-source field profile, which agrees well with the φ enics benchmark for the same $\mathcal{O}(1)$ parameter set, alongside the loss as a function of the training epoch. Although the loss fluctuates substantially, a downward trend in the floor value is visible, confirming that training is progressing. Overall the results indicate that whilst **Elvet** can successfully solve the linear KG equation, the full IR Galileon system poses a substantially harder optimisation problem.

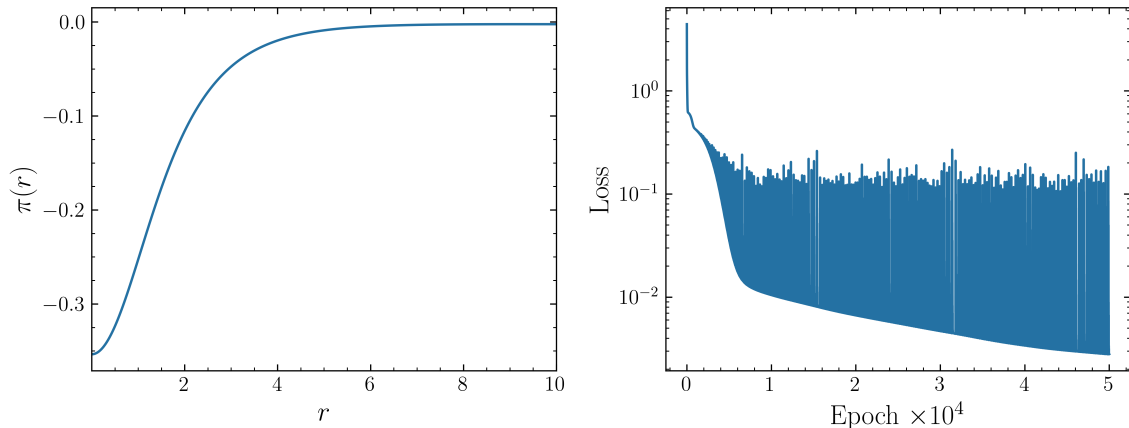


Figure 5: The left panel shows **Elvet**’s field solution around the source for the $\mathcal{O}(1)$ parameter set which matches the comparable result from φ enics. The right plot shows the loss as a function of the epochs during training. The loss does decrease throughout training however large inter-epoch fluctuations are clearly visible.

5 Discussion & Outlook

This project pursued two primary objectives, both motivated by the limitations encountered in semester 7 [27]. The first was to obtain converged numerical solutions of the IR Galileon equation of motion with both $n = 3$ and $n = 4$ nonlinear operators present simultaneously – a task that proved intractable via direct Newton-Raphson methods in φ enics [26]. The second was to establish an independent numerical benchmark using the **Elvet** [29] neural network solver. This offered a qualitatively different approach to the problem and could in principle be more straightforwardly extended to more complex operator combinations than φ enics, which was specifically designed for this class of equations.

The continuation approach successfully addressed the first objective. By gradually introducing the $n = 4$ operator through a homotopy parameter, converged solutions of the combined system were obtained in parameter regimes where direct solution had previously failed. The results indicate that the higher-order operator produces qualitative rather than perturbative modifications to the field profile: the Vainshtein radius is shifted outward by more than an order of magnitude, and a two-stage power-law structure emerges in the transitional regime, suggestive of competitive dynamics between the two operators at different scales. These findings are in tension with the effective field theory expectation that higher-dimension operators should remain subdominant below the cutoff scale. While some numerical noise persists – with sensitivity to step size and internal scaling parameters – the solutions represent a substantial improvement over the semester 7 results and provide the first indication of how the combined system behaves in the Vainshtein regime. Further consideration of these results is presented in [47].

The second objective was only partially met. **Elvet** performed well as an independent solver for the linear Klein-Gordon equation, producing solutions in excellent agreement with Mathematica benchmarks and demonstrating its validity as a cross-check tool in principle. However, the full nonlinear infra-red Galileon system proved significantly more challenging. The combination of nested nonlinear operators, a large hierarchy of physical scales, and exponentially suppressed asymptotic behaviour produced a loss landscape in which gradient-based optimisation struggled to find globally physical solutions. Near-source field profiles consistent with φ enics were obtained. This suggests that the primary optimisation difficulty lies not within the screened region itself, but in reproducing the large- r behaviour. Numerous modifications – architecture, domain points and spacing, boundary conditions, loss reweighting, domain decomposition – were explored without resolving this. Many of these modifications extend substantially beyond the standard workflow envisioned by the framework.

While a mesh-free neural network solver is an appealing alternative to a finite-element code tuned to a specific problem, the practical difficulties encountered here suggest that strongly nonlinear systems with large scale separation may require deeper modifications to the framework before reliable solutions can be obtained. These might include a physically motivated ansatz from which the network can train on deviations from, adaptive weighting, or curriculum-based approaches that first learn the linear solution before introducing nonlinear terms progressively. Such developments could make **Elvet** a genuinely useful complement to φ enics, and represent a natural direction for future work.

In summary, continuation methods successfully produced converged solutions of the combined $n = 3$, $n = 4$ Galileon system in regimes where direct approaches failed, providing evidence that higher-order operators can significantly alter the screened dynamics. This challenges the naive EFT expectation that such operators remain perturbatively suppressed within the Vainshtein regime. In parallel, the investigation of **Elvet** demonstrated both the promise and present limitations of neural network differential equation solvers for strongly nonlinear field theories with widely separated scales. Together, these results highlight the importance of robust numerical methodology in probing the structure of modified gravity effective field theories.

Acknowledgments

The work carried out in this project was inspired by the φ enics paper [26] and uses the software [25] and [29]. The formatting of this report is primarily based upon the JHEP LaTeX format, with some minor modifications; the original files can be accessed [here](#). The author would also like to recognise the helpful insight provided by both Christoph Englert and Peter Millington in their supervision of this project.

References

1. Chandrasekhar, S. The General Theory of Relativity: Why “It Is Probably the Most Beautiful of All Existing Theories”. *Journal of Astrophysics and Astronomy* **5**, 3–11. ISSN: 0250-6335. <https://ui.adsabs.harvard.edu/abs/1984JApA...5....3C> (2025) (1st Mar. 1984).
2. Hawking, S. W. Black Holes and Thermodynamics. *Physical Review D* **13**, 191–197. <https://link.aps.org/doi/10.1103/PhysRevD.13.191> (2026) (15th Jan. 1976).
3. Baumann, D. *Cosmology* ISBN: 978-1-108-93709-2 978-1-108-83807-8 (Cambridge University Press, July 2022).
4. Shapiro, S. S., Davis, J. L., Lebach, D. E. & Gregory, J. S. Measurement of the Solar Gravitational Deflection of Radio Waves Using Geodetic Very-Long-Baseline Interferometry Data, 1979–1999. *Physical Review Letters* **92**, 121101. <https://link.aps.org/doi/10.1103/PhysRevLett.92.121101> (2025) (26th Mar. 2004).
5. Clifton, T., Ferreira, P. G., Padilla, A. & Skordis, C. Modified Gravity and Cosmology. *Physics Reports. Modified Gravity and Cosmology* **513**, 1–189. ISSN: 0370-1573. <https://www.sciencedirect.com/science/article/pii/S0370157312000105> (2025) (1st Mar. 2012).
6. Einstein, A. The Foundation of the General Theory of Relativity. *Annalen Phys.* **49**, 769–822 (1916).
7. Dyson, F. W., Eddington, A. S. & Davidson, C. A Determination of the Deflection of Light by the Sun’s Gravitational Field, from Observations Made at the Total Eclipse of May 29, 1919. *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character* **220**, 291–333. ISSN: 0264-3952. JSTOR: 91137. <https://www.jstor.org/stable/91137> (2025) (1920).
8. Will, C. M. The 1919 Measurement of the Deflection of Light. *Classical and Quantum Gravity* **32**, 124001. ISSN: 0264-9381. <https://doi.org/10.1088/0264-9381/32/12/124001> (2025) (June 2015).
9. Shapiro, I. I. Fourth Test of General Relativity. *Physical Review Letters* **13**, 789–791. <https://link.aps.org/doi/10.1103/PhysRevLett.13.789> (2025) (28th Dec. 1964).
10. Bertotti, B., Iess, L. & Tortora, P. A Test of General Relativity Using Radio Links with the Cassini Spacecraft. *Nature* **425**, 374–376. ISSN: 1476-4687. <https://www.nature.com/articles/nature01997> (2025) (Sept. 2003).
11. LIGO Scientific Collaboration and Virgo Collaboration *et al.* Observation of Gravitational Waves from a Binary Black Hole Merger. *Physical Review Letters* **116**, 061102. <https://link.aps.org/doi/10.1103/PhysRevLett.116.061102> (2026) (11th Feb. 2016).
12. Carroll, S. *Spacetime and Geometry: An Introduction to General Relativity* xiv+513. ISBN: 978-0-8053-8732-2 (Benjamin Cummings, San Francisco, Calif. ; 2004).

13. Schwartz, M. D. *Quantum Field Theory and the Standard Model* ISBN: 978-1-108-98503-1 (Cambridge University Press, 15th Dec. 2013).
14. Will, C. M. *Theory and Experiment in Gravitational Physics* Second edition. 350 pp. ISBN: 978-1-316-33861-2 (University Press, Cambridge, 2018).
15. Kaluza, Th. Zum Unitätsproblem Der Physik. *Sitzungsber. Preuss. Akad. Wiss. Berlin (Math. Phys.)* **1921**, 966–972 (1921).
16. Klein, O. Quantentheorie und fünfdimensionale Relativitätstheorie. *Zeitschrift für Physik* **37**, 895–906. ISSN: 0044-3328. <https://doi.org/10.1007/BF01397481> (2025) (1st Dec. 1926).
17. (a) Randall, L. & Sundrum, R. Large Mass Hierarchy from a Small Extra Dimension. *Physical Review Letters* **83**, 3370–3373. <https://link.aps.org/doi/10.1103/PhysRevLett.83.3370> (2025) (25th Oct. 1999); (b) Randall, L. & Sundrum, R. An Alternative to Compactification. *Physical Review Letters* **83**, 4690–4693. <https://link.aps.org/doi/10.1103/PhysRevLett.83.4690> (2025) (6th Dec. 1999).
18. Dvali, G., Gabadadze, G. & Porrati, M. 4D Gravity on a Brane in 5D Minkowski Space. *Physics Letters B* **485**, 208–214. ISSN: 0370-2693. <https://www.sciencedirect.com/science/article/pii/S0370269300006699> (2025) (13th July 2000).
19. Brans, C. & Dicke, R. H. Mach’s Principle and a Relativistic Theory of Gravitation. *Physical Review* **124**, 925–935. <https://link.aps.org/doi/10.1103/PhysRev.124.925> (2025) (1st Nov. 1961).
20. Carroll, S. M. The Cosmological Constant. *Living Reviews in Relativity* **4**, 1. ISSN: 1433-8351. <https://doi.org/10.12942/lrr-2001-1> (2025) (7th Feb. 2001).
21. Weinberg, S. The Cosmological Constant Problem. *Reviews of Modern Physics* **61**, 1–23. <https://link.aps.org/doi/10.1103/RevModPhys.61.1> (2025) (1st Jan. 1989).
22. Joyce, A., Jain, B., Khoury, J. & Trodden, M. Beyond the Cosmological Standard Model. *Physics Reports* **568**, 1–98. ISSN: 0370-1573. <https://www.sciencedirect.com/science/article/pii/S0370157314004487> (2025) (22nd Mar. 2015).
23. Khoury, J. & Weltman, A. Chameleon Cosmology. *Physical Review D* **69**, 044026. <https://link.aps.org/doi/10.1103/PhysRevD.69.044026> (2025) (27th Feb. 2004).
24. Nicolis, A., Rattazzi, R. & Trincherini, E. Galileon as a Local Modification of Gravity. *Physical Review D* **79**, 064036. <https://link.aps.org/doi/10.1103/PhysRevD.79.064036> (2025) (31st Mar. 2009).
25. Saadeh, D. *Phienics* 6th July 21. <https://github.com/scaramouche-00/phienics>.
26. Braden, J., Burrage, C., Elder, B. & Saadeh, D. φ enics: Vainshtein Screening with the Finite Element Method. *Journal of Cosmology and Astroparticle Physics* **2021**, 010. ISSN: 1475-7516. <https://doi.org/10.1088/1475-7516/2021/03/010> (2025) (Mar. 2021).

27. Broxton, T. *Cracking the Cosmos* MPhys Report (University of Manchester, 18th Dec. 2025). 20 pp. <https://theobroxton.github.io/Sem%207%20Report.pdf>.
28. Allgower, E. L. *Introduction to Numerical Continuation Methods Classics in Applied Mathematics* **45**. ISBN: 978-0-89871-915-4 (Society for Industrial and Applied Mathematics SIAM Market Street, Floor 6, Philadelphia, PA 19104, Philadelphia, Pa, 2003).
29. Araz, J. Y., Criado, J. C. & Spannowsky, M. *Elvet – a Neural Network-Based Differential Equation and Variational Problem Solver* arXiv: [2103.14575 \[cs\]](https://arxiv.org/abs/2103.14575). <http://arxiv.org/abs/2103.14575> (2026). Pre-published.
30. Woodard, R. P. *The Theorem of Ostrogradsky* arXiv: [1506.02210 \[hep-th\]](https://arxiv.org/abs/1506.02210). <http://arxiv.org/abs/1506.02210> (2025). Pre-published.
31. Rham, C. de. Galileons in the sky. *Comptes Rendus. Physique* **13**, 666–681. ISSN: 1878-1535. <https://comptes-rendus.academie-sciences.fr/physique/articles/10.1016/j.crhy.2012.04.006/> (2025) (2012).
32. De Rham, C., Melville, S., Tolley, A. J. & Zhou, S.-Y. Massive Galileon Positivity Bounds. *Journal of High Energy Physics* **2017**, 72. ISSN: 1029-8479. [https://doi.org/10.1007/JHEP09\(2017\)072](https://doi.org/10.1007/JHEP09(2017)072) (2025) (18th Sept. 2017).
33. Burrage, C., Coltman, B., Padilla, A., Saadeh, D. & Wilson, T. Massive Galileons and Vainshtein Screening. *Journal of Cosmology and Astroparticle Physics* **2021**, 050–050. ISSN: 1475-7516. arXiv: [2008.01456 \[hep-th\]](https://arxiv.org/abs/2008.01456). <http://arxiv.org/abs/2008.01456> (2025) (23rd Feb. 2021).
34. *Numerical Continuation Methods for Dynamical Systems: Path Following and Boundary Value Problems* (eds Krauskopf, B., Osinga, H. M. & Galán-Vioque, J.) ISBN: 978-1-4020-6355-8 978-1-4020-6356-5. <http://link.springer.com/10.1007/978-1-4020-6356-5> (2026) (Springer Netherlands, Dordrecht, 2007).
35. Margenberg, N., Hartmann, D., Lessig, C. & Richter, T. A Neural Network Multigrid Solver for the Navier-Stokes Equations. *Journal of Computational Physics* **460**, 110983. ISSN: 00219991. arXiv: [2008.11520 \[physics.comp-ph\]](https://arxiv.org/abs/2008.11520). <http://arxiv.org/abs/2008.11520> (2026) (July 2022).
36. *The Nobel Prize in Physics 2024* NobelPrize.org. <https://www.nobelprize.org/prizes/physics/2024/summary/> (2026).
37. Latz, J. *Mathematics and Applications of Machine Learning* Lecture Notes. 21st Nov. 2025.
38. *Neural Networks – TikZ.Net* https://tikz.net/neural_networks/ (2026).
39. Hornik, K., Stinchcombe, M. & White, H. Multilayer Feedforward Networks Are Universal Approximators. *Neural Networks* **2**, 359–366. ISSN: 0893-6080. <https://www.sciencedirect.com/science/article/pii/0893608089900208> (2026) (1st Jan. 1989).

40. Pinkus, A. Approximation Theory of the MLP Model in Neural Networks. *Acta Numerica* **8**, 143–195. ISSN: 1474-0508, 0962-4929. <https://www.cambridge.org/core/journals/acta-numerica/article/approximation-theory-of-the-mlp-model-in-neural-networks/18072C558C8410C4F92A82BCC8FC8CF9> (2026) (Jan. 1999).
41. Telgarsky, M. *Benefits of Depth in Neural Networks* in *Conference on Learning Theory* Conference on Learning Theory (PMLR, 6th June 2016), 1517–1539. <https://proceedings.mlr.press/v49/telgarsky16.html> (2026).
42. Higham, C. F. & Higham, D. J. Deep Learning: An Introduction for Applied Mathematicians. *SIAM Review* **61**, 860–891. ISSN: 0036-1445. JSTOR: [45476479](https://www.jstor.org/stable/45476479). <https://www.jstor.org/stable/45476479> (2026) (2019).
43. Kingma, D. P. & Ba, J. *Adam: A Method for Stochastic Optimization* arXiv: [1412.6980](https://arxiv.org/abs/1412.6980) [cs.LG]. <http://arxiv.org/abs/1412.6980> (2026). Pre-published.
44. Bottou, L., Curtis, F. E. & Nocedal, J. Optimization Methods for Large-Scale Machine Learning. *SIAM Review* **60**, 223–311. ISSN: 0036-1445. JSTOR: [45109416](https://www.jstor.org/stable/45109416). <https://www.jstor.org/stable/45109416> (2026) (2018).
45. Abadi, M. *et al.* *TensorFlow: A System for Large-Scale Machine Learning* arXiv: [1605.08695](https://arxiv.org/abs/1605.08695) [cs.DC]. <http://arxiv.org/abs/1605.08695> (2026). Pre-published.
46. Araz, J. Y., Criado, J. C. & Spannowsky, M. *Index — Elvet Documentation* <https://elvet.gitlab.io/elvet/index.html> (2026).
47. Wright, E. *Improved Numerics and Analytic Benchmarks for Simulations of Vainshtein Screening in the Presence of Higher-Order Terms* MPhys Report (University of Manchester, 14th May 26).